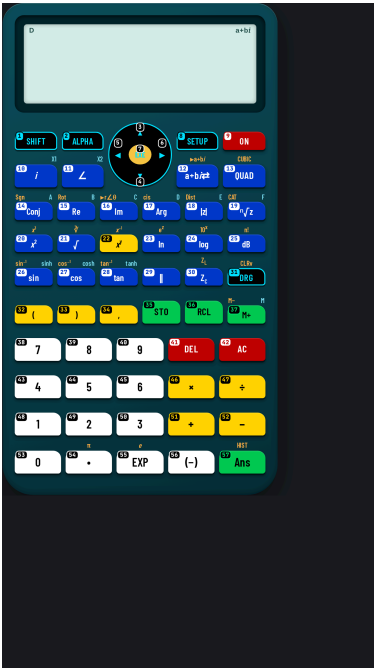
U-45 CALCULATORS

User Manual

Calculator #1 Complex-Number Engineering Scientific (teal)

Scientific – natural display



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#1 — Complex-Number Engineering Scientific (teal)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 SHIFT]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

algebraic complex-number entry (type a+bi or r$\angle\theta$, then EXE).

Chapters

1. Getting started
2. Features
3. Feature guide (2b)
4. Numbered key map
5. Key-by-key reference
6. Worked examples
7. Tricks & hacks
8. Keyboard shortcuts
9. Common mistakes
10. Practice drills
11. Appendix A — Drill answers

12. Appendix B — Function finder (A–Z)

13. Appendix C — Glossary

14. Appendix D — Model notes & limits

15. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#1** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#38, #39, #40, #43, #44, #45, #48, #49, #50, #53, #54, #55, #56
op	Operators & execute	#7, #22, #32, #33, #34, #46, #47, #51, #52
fn	Functions	#10, #11, #12, #13, #14, #15, #16, #17, #18, #19, #20, #21, #23, #24, #25, #26, #27, #28, #29, #30
mem	Memory, variables & registers	#35, #36, #37, #57
clr	Clear, delete & power	#9, #41, #42

mod	**Shift, modes, menus & navigation**	**#1, #2, #3, #4, #5, #6, #8, #31**

2 · Features

49 features: 35 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Imaginary unit i key: type complex numbers as a+bi (e.g. 3+4i)**
2. LIVE **Polar entry r∠θ with the ∠ key (θ in the current angle unit)**
3. LIVE **Complex addition, subtraction, multiplication and division**
4. LIVE **Complex powers z^w and integer powers**
5. LIVE **Square root of negative and complex numbers ($\sqrt{-4} = 2i$)**
6. LIVE **Complex exponential e^z and natural log ln z (principal value)**
7. LIVE **Complex sin, cos, tan and hyperbolic functions**
8. LIVE **Real part Re(z)**
9. LIVE **Imaginary part Im(z)**
10. LIVE **Complex conjugate Conj(z)**
11. LIVE **Argument Arg(z) in degrees, radians or grads**
12. LIVE **Modulus |z|**
13. LIVE **All n complex n-th roots ⁿRoots(z,n) listed k = 0…n−1**
14. LIVE **Result-form cycle a+bi → r∠θ → r·e^(θi) with one key (re-formats the last answer)**
15. LIVE **Direct ►a+bi and ►r∠θ display keys**
16. LIVE **cis(θ) unit phasor**
17. LIVE **Rotate a phasor: Rot(z,θ)**
18. LIVE **Unit phasor Sgn(z) = z/|z|**
19. LIVE **Parallel-impedance operator a‖b = ab/(a+b)**

20. LIVE **Capacitor impedance Zc(C,f) = $-i/(2\pi fC)$**
21. LIVE **Inductor impedance ZL(L,f) = $2\pi fL \cdot i$**
22. LIVE **Gain in decibels dB(z) = $20 \cdot \log_{10}|z|$**
23. LIVE **Distance between two complex points Dist(a,b)**
24. LIVE **Quadratic solver with complex roots (stores X1, X2)**
25. LIVE **Cubic solver with complex roots (stores X1…X3)**
26. LIVE **Variables A–F, X1, X2 and M with STO / RCL**
27. LIVE **M+ / M− running memory**
28. LIVE **Two-line history with ▲ ▼ recall and editing**
29. LIVE **Cursor editing with ◀ ▶ and DEL**
30. LIVE **Angle-unit cycle D → R → G (DRG)**
31. LIVE **SETUP menu: angle units and Fix / Sci / Eng / Normal formats**
32. LIVE **Catalogue menu of every complex function**
33. LIVE **Auto-closing brackets on EXE**
34. LIVE **Ans chaining (an operator first continues from Ans)**
35. LIVE **π, e, x^2, x^3, x^{-1}, $\sqrt[3]{}$, log, 10^x and n! on the keys**
36. APP/UI **Every key numbered 1…N in reading order — matches the key map in the user manual**
37. APP/UI **High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a "Classic" toggle**
38. APP/UI **Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)**
39. APP/UI **Show / hide key numbers with one button (remembered between sessions)**

40. APP/UI **Copy the current display text to the clipboard**
41. APP/UI **Physical-keyboard shortcuts: digits, + − * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i**
42. APP/UI **Fit / 1:1 zoom that never pushes keys off a phone screen**
43. APP/UI **Reference photo viewer and tool-tip name on every key**
44. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
45. PLANNED **Exam mode (locks memory & shows indicator)**
46. PLANNED **Answer history export to text**
47. PLANNED **Speech read-out of results**
48. PLANNED **Colour-blind safe key theme**
49. PLANNED **Custom shortcut keys**

2b · Feature guide

Calculator #1 is a algebraic complex-number entry (type a+bi or r∠θ, then EXE). Each card below is one of its 35 working features, with the keys that reach it and, where one exists, a verified example.

F1 Imaginary unit i key: type complex numbers as a+bi (e.g. 3+4i)

Where: [#10 i] Imaginary unit i; [#14 Conj] Complex conjugate; [#8 SETUP] SETUP — angle unit, number format, display options

See it in action: Multiply two complex numbers (real part) — $Re((2+3i)(4-i))$ = 11; Modulus of a complex number — $|3+4i|$ = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F2 Polar entry r∠θ with the ∠ key (θ in the current angle unit)

Where: [#17 Arg] Argument (angle) of z in the current unit; [#26 sin] Sine (uses the current angle unit); [#27 cos] Cosine (current angle unit)

See it in action: Polar to rectangular (real part) — $Re(2\angle 60°)$ = 1

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F3 Complex addition, subtraction, multiplication and division

Where: #13 QUAD Quadratic equation solver (complex roots); #14 Conj Complex conjugate; #18 |z| Modulus |z|

See it in action: Multiply two complex numbers (real part) — *Re((2+3i)(4−i))* = 11; Modulus of a complex number — *|3+4i|* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F4 Complex powers z^w and integer powers

Where: #13 QUAD Quadratic equation solver (complex roots); #14 Conj Complex conjugate; #18 |z| Modulus |z|

See it in action: Multiply two complex numbers (real part) — *Re((2+3i)(4−i))* = 11; Modulus of a complex number — *|3+4i|* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F5 Square root of negative and complex numbers (√−4 = 2*i*)

Where: #21 √ Square root √; #13 QUAD Quadratic equation solver (complex roots); #14 Conj Complex conjugate

See it in action: Multiply two complex numbers (real part) — *Re((2+3i)(4−i))* = 11; Modulus of a complex number — *|3+4i|* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F6 Complex exponential e^z and natural log ln z (principal value)

Where: #23 ln Natural logarithm ln; #12 a+bi⇄ Cycle result form: rectangular → polar → exponential; #13 QUAD Quadratic equation solver (complex roots)

See it in action: Multiply two complex numbers (real part) — *Re((2+3i)(4−i))* = 11; Modulus of a complex number — *|3+4i|* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F7 Complex sin, cos, tan and hyperbolic functions

Where: #17 Arg Argument (angle) of z in the current unit; #19 $\sqrt[n]{z}$ All n complex n-th roots: ^{n}Roots(z,n); #13 QUAD Quadratic equation solver (complex roots)

See it in action: Multiply two complex numbers (real part) — *Re((2+3i)(4−i))* = 11; Modulus of a complex number — *|3+4i|* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F8 Real part Re(z)

Where: #15 Re Real part; #16 Im Imaginary part

See it in action: Multiply two complex numbers (real part) — *Re((2+3i)(4−i))* = 11; Imaginary part of a quotient — *Im((1+i)÷(1−i))* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F9 Imaginary part Im(z)

Where: #16 Im Imaginary part; #10 i Imaginary unit i; #15 Re Real part

See it in action: Multiply two complex numbers (real part) — *Re((2+3i)(4−i))* = 11; Imaginary part of a quotient — *Im((1+i)÷(1−i))* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F10 Complex conjugate Conj(z)

Where: #14 Conj Complex conjugate; #13 QUAD Quadratic equation solver (complex roots); #18 |z| Modulus |z|

See it in action: Multiply two complex numbers (real part) — *Re((2+3i)(4−i))* = 11; Modulus of a complex number — *|3+4i|* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F11 Argument Arg(z) in degrees, radians or grads

Where: #17 Arg Argument (angle) of z in the current unit; #34 , Comma / argument separator (between function arguments, list items or coordinates)

See it in action: Argument in degrees — *Arg(1+i)* = 45

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F12 Modulus |z|

Where: #18 |z| Modulus |z|

See it in action: Modulus of a complex number — *|3+4i|* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F13 All n complex n-th roots ⁿRoots(z,n) listed k = 0…n−1

Where: #13 QUAD Quadratic equation solver (complex roots); #19 ⁿ√z All n complex n-th roots: ⁿRoots(z,n); #14 Conj Complex conjugate

See it in action: Multiply two complex numbers (real part) — *Re((2+3i)(4−i))* = 11; Modulus of a complex number — *|3+4i|* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F14 Result-form cycle a+b*i* → r∠θ → r·e^(θ*i*) with one key (re-formats the last answer)

Where: #12 a+bi⇄ Cycle result form: rectangular → polar → exponential; #57 Ans Ans — recall the last answer into the current expression; #7 EXE Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F15 Direct ▶a+b*i* and ▶r∠θ display keys

Where: #8 SETUP SETUP — angle unit, number format, display options

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F16 cis(θ) unit phasor

Where: #14 Conj Complex conjugate; #17 Arg Argument (angle) of z in the current unit; #8 SETUP SETUP — angle unit, number format, display options

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F17 Rotate a phasor: Rot(z,θ)

Where: #15 Re Real part; #14 Conj Complex conjugate

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F18 Unit phasor Sgn(z) = z/|z|

Where: #14 Conj Complex conjugate; #8 SETUP SETUP — angle unit, number format, display options; #10 i Imaginary unit i

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F19 Parallel-impedance operator a‖b = ab/(a+b)

Where: #29 ‖ Parallel combination a‖b = ab/(a+b); #30 ZC Capacitor impedance Zc(C,f)

See it in action: Parallel resistors — *6 ‖ 3 (Ω)* = 2

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F20 Capacitor impedance Zc(C,f) = −i/(2πfC)

Where: #30 ZC Capacitor impedance Zc(C,f)

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F21 Inductor impedance ZL(L,f) = 2πfL·*i*

Where: #30 ZC **Capacitor impedance Zc(C,f)**

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F22 Gain in decibels dB(z) = 20·$\log_{10}|z|$

Where: #25 dB **Gain in decibels 20·log|z|;** #24 log **Common logarithm $\log_{10}$**

See it in action: Gain in dB — *dB(10)* = 20

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F23 Distance between two complex points Dist(a,b)

Where: #18 |z| Modulus |z|; #13 QUAD Quadratic equation solver (complex roots); #14 Conj Complex conjugate

See it in action: Multiply two complex numbers (real part) — *Re((2+3i)(4−i))* = 11; Modulus of a complex number — *|3+4i|* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F24 Quadratic solver with complex roots (stores X1, X2)

Where: #13 QUAD Quadratic equation solver (complex roots); #19 $\sqrt[n]{z}$ All n complex n-th roots: ^{n}Roots(z,n); #14 Conj Complex conjugate

See it in action: Multiply two complex numbers (real part) — *Re((2+3i)(4−i))* = 11; Modulus of a complex number — *|3+4i|* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F25 Cubic solver with complex roots (stores X1…X3)

Where: #13 QUAD Quadratic equation solver (complex roots); #19 $\sqrt[n]{z}$ All n complex n-th roots: ^{n}Roots(z,n); #14 Conj Complex conjugate

See it in action: Multiply two complex numbers (real part) — *Re((2+3i)(4−i))* = 11; Modulus of a complex number — *|3+4i|* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F26 Variables A–F, X1, X2 and M with STO / RCL

Where: #31 DRG DRG — choose/convert the angle unit (Deg / Rad / Grad); #35 STO STO — store the value into a memory / variable (press the target next); #36 RCL RCL — recall a memory / variable

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F27 M+ / M− running memory

Where: #35 STO STO — store the value into a memory / variable (press the target next); #36 RCL RCL — recall a memory / variable; #37 M+ M+ — add the value to independent memory M

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F28 Two-line history with ▲ ▼ recall and editing

Where: #3 ▲ Cursor ▲ — up / previous history line / previous item; #4 ▼ Cursor ▼ — down / next history line / next item; #57 Ans Ans — recall the last answer into the current expression

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F29 Cursor editing with ◀▶ and DEL

Where: #41 DEL DEL — delete the character at the cursor; #3 ▲ Cursor ▲ — up / previous history line / previous item; #4 ▼ Cursor ▼ — down / next history line / next item

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F30 Angle-unit cycle D → R → G (DRG)

Where: #31 DRG DRG — choose/convert the angle unit (Deg / Rad / Grad); #8 SETUP SETUP — angle unit, number format, display options; #17 Arg Argument (angle) of z in the current unit

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F31 SETUP menu: angle units and Fix / Sci / Eng / Normal formats

Where: #8 SETUP SETUP — angle unit, number format, display options; #15 Re Real part; #17 Arg Argument (angle) of z in the current unit

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F32 Catalogue menu of every complex function

Where: #19 $\sqrt[n]{z}$ All n complex n-th roots: ⁿRoots(z,n); #1 SHIFT SHIFT / 2nd — the next key uses its upper (shift-coloured) function; #13 QUAD Quadratic equation solver (complex roots)

See it in action: Multiply two complex numbers (real part) — *Re((2+3i)(4−i))* = 11; Modulus of a complex number — *|3+4i|* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F33 Auto-closing brackets on EXE

Where: #33) Closing bracket). Closing brackets at the end of a line may be omitted; #7 EXE Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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F34 Ans chaining (an operator first continues from Ans)

Where: #57 Ans Ans — recall the last answer into the current expression; #54 • Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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_____

F35 π, e, x^2, x^3, x^{-1}, $\sqrt[3]{}$, log, 10^x and n! on the keys

Where: **#24 log** Common logarithm $\log_{10}$; **#25 dB** Gain in decibels 20·log|z|

Try it: Press Reset, use the key(s) above, and write the display here:

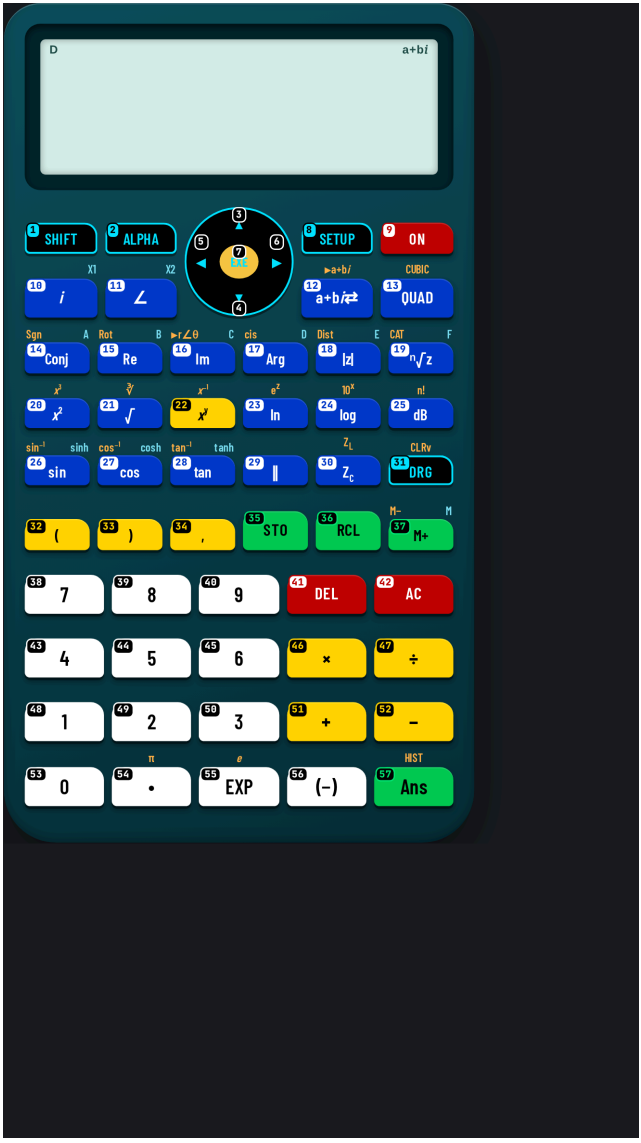
_____

Your notes:

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3 · Numbered key map



Calculator #1 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	SHIFT	·	·	·	mod
2	ALPHA	·	·	·	mod
3	▲	·	·	·	mod
4	▼	·	·	·	mod
5	◀	·	·	·	mod
6	▶	·	·	·	mod
7	EXE	·	·	·	op
8	SETUP	·	·	·	mod
9	ON	·	·	·	clr
10	i	·	X1	·	fn
11	∠	·	X2	·	fn
12	a+bi⇄	▸a+bi	·	·	fn
13	QUAD	CUBIC	·	·	fn
14	Conj	Sgn	A	·	fn
15	Re	Rot	B	·	fn
16	Im	▸r∠θ	C	·	fn
17	Arg	cis	D	·	fn
18	|z|	Dist	E	·	fn
19	$\sqrt[n]{z}$	CAT	F	·	fn
20	x^2	x^3	·	·	fn
21	$\sqrt{}$	$\sqrt[3]{}$	·	·	fn
22	xy	x^{-1}	·	·	op
23	ln	ez	·	·	fn

#	Face	Shift	Alpha	3rd	Group
24	log	10x	·	·	fn
25	dB	n!	·	·	fn
26	sin	$\sin^{-1}$	sinh	·	fn
27	cos	$\cos^{-1}$	cosh	·	fn
28	tan	$\tan^{-1}$	tanh	·	fn
29	‖	·	·	·	fn
30	ZC	ZL	·	·	fn
31	DRG	CLRv	·	·	mod
32	(	·	·	·	op
33	)	·	·	·	op
34	,	·	·	·	op
35	STO	·	·	·	mem
36	RCL	·	·	·	mem
37	M+	M−	M	·	mem
38	7	·	·	·	num
39	8	·	·	·	num
40	9	·	·	·	num
41	DEL	·	·	·	clr
42	AC	·	·	·	clr
43	4	·	·	·	num
44	5	·	·	·	num
45	6	·	·	·	num
46	×	·	·	·	op
47	÷	·	·	·	op
48	1	·	·	·	num
49	2	·	·	·	num
50	3	·	·	·	num
51	+	·	·	·	op

#	Face	Shift	Alpha	3rd	Group
52	−	·	·	·	op
53	0	·	·	·	num
54	•	π	·	·	num
55	EXP	e	·	·	num
56	(−)	·	·	·	num
57	Ans	HIST	·	·	mem

4 · Key-by-key reference

Every key on calculator #1, in number order, with every function it carries in this app.

#1 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#2 ALPHA `Shift, modes, menus & navigation`

Primary: ALPHA — the next key types its letter/variable (red/green label)

#3 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#4 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#5 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

#6 ▶ `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

#7 EXE `Operators & execute`

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

#8 SETUP `Shift, modes, menus & navigation`

Primary: SETUP — angle unit, number format, display options

#9 ON `Clear, delete & power`

Primary: ON — power on

#10 i `Functions`

Primary: Imaginary unit i

alpha layer (press #2 first): X1 — Variable X1

#11 ∠ `Functions`

Primary: Polar entry r∠θ

alpha layer (press #2 first): X2 — Variable X2

#12 a+bi⇄ `Functions`

Primary: Cycle result form: rectangular → polar → exponential

shifted (2nd / f / SHIFT layer) (press #1 first): ▸a+bi — Show results in rectangular form

#13 QUAD Functions

Primary: Quadratic equation solver (complex roots)

shifted (2nd / f / SHIFT layer) (press #1 first): CUBIC — Cubic equation solver (complex roots)

#14 Conj Functions

Primary: Complex conjugate

shifted (2nd / f / SHIFT layer) (press #1 first): Sgn — Unit phasor z/|z|

alpha layer (press #2 first): A — Variable A (store with STO/ →, recall by name)

#15 Re Functions

Primary: Real part

shifted (2nd / f / SHIFT layer) (press #1 first): Rot — Rotate z by angle θ: Rot(z,θ)

alpha layer (press #2 first): B — Variable B (store with STO/ →, recall by name)

#16 Im Functions

Primary: Imaginary part

shifted (2nd / f / SHIFT layer) (press #1 first): ▸r∠θ — Show results in polar form

alpha layer (press #2 first): C — Variable C (store with STO/ →, recall by name)

#17 Arg Functions

Primary: Argument (angle) of z in the current unit

shifted (2nd / f / SHIFT layer) (press #1 first): cis — cis(θ) = cos θ + i sin θ

alpha layer (press #2 first): D — Variable D (store with STO/ →, recall by name)

#18 |z| Functions

Primary: Modulus |z|

shifted (2nd / f / SHIFT layer) (press #1 first): Dist — Distance between two complex points

alpha layer (press #2 first): E — Variable E (store with STO/ →, recall by name)

#19 $\sqrt[n]{z}$ Functions

Primary: All n complex n-th roots: ^{n}Roots(z,n)

shifted (2nd / f / SHIFT layer) (press #1 first): CAT — Catalogue of complex functions

alpha layer (press #2 first): F — Variable F (store with STO/ →, recall by name)

#20 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #1 first): x^3 — Cube x^3

Keyboard: q

#21 √ Functions

Primary: Square root √

shifted (2nd / f / SHIFT layer) (press #1 first): $\sqrt[3]{}$ — Cube root $\sqrt[3]{}$

Keyboard: r

#22 xy Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

shifted (2nd / f / SHIFT layer) (press #1 first): x^{-1} — Reciprocal x^{-1} (1/x)

Keyboard: ^

#23 ln Functions

Primary: Natural logarithm ln

shifted (2nd / f / SHIFT layer) (press #1 first): ez — e^x — exponential

Keyboard: n

#24 log Functions

Primary: Common logarithm $\log_{10}$

shifted (2nd / f / SHIFT layer) (press #1 first): 10x — 10^x — antilog

Keyboard: l

#25 dB Functions

Primary: Gain in decibels 20·log|z|

shifted (2nd / f / SHIFT layer) (press #1 first): n! — Factorial n! (also on non-integers via the gamma function where supported)

#26 sin Functions

Primary: Sine (uses the current angle unit)

shifted (2nd / f / SHIFT layer) (press #1 first): $\sin^{-1}$ — Inverse sine $\sin^{-1}$

alpha layer (press #2 first): sinh — sinh

Keyboard: s

#27 cos Functions

Primary: Cosine (current angle unit)

shifted (2nd / f / SHIFT layer) (press #1 first): $\cos^{-1}$ — Inverse cosine $\cos^{-1}$

alpha layer (press #2 first): cosh — cosh

Keyboard: c

#28 tan Functions

Primary: Tangent (current angle unit)

shifted (2nd / f / SHIFT layer) (press #1 first): $\tan^{-1}$ — Inverse tangent $\tan^{-1}$

alpha layer (press #2 first): tanh — tanh

Keyboard: t

#29 ‖ Functions

Primary: Parallel combination a‖b = ab/(a+b)

#30 ZC Functions

Primary: Capacitor impedance Zc(C,f)

shifted (2nd / f / SHIFT layer) (press #1 first): ZL — Inductor impedance ZL(L,f)

#31 DRG Shift, modes, menus & navigation

Primary: DRG — choose/convert the angle unit (Deg / Rad / Grad)

shifted (2nd / f / SHIFT layer) (press #1 first): CLRv — Clear variables

#32 (Operators & execute

Primary: Opening bracket (

Keyboard: (

#33) Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

Keyboard:)

#34 , Operators & execute

Primary: Comma / argument separator (between function arguments, list items or coordinates)

Keyboard: ,

#35 STO Memory, variables & registers

Primary: STO — store the value into a memory / variable (press the target next)

#36 RCL Memory, variables & registers

Primary: RCL — recall a memory / variable

#37 M+ Memory, variables & registers

Primary: M+ — add the value to independent memory M

shifted (2nd / f / SHIFT layer) (press #1 first): M− — M− — subtract the value from memory M

alpha layer (press #2 first): M — Variable M (store with STO/ →, recall by name)

#38 7 Digits & number entry

Primary: Digit 7

Keyboard: 7

#39 8 Digits & number entry

Primary: Digit 8

Keyboard: 8

#40 9 Digits & number entry

Primary: Digit 9

Keyboard: 9

#41 DEL `Clear, delete & power`

Primary: DEL — delete the character at the cursor

Keyboard: `Backspace` `Delete`

#42 AC `Clear, delete & power`

Primary: AC — clear all of the current calculation

Keyboard: `Escape`

#43 4 `Digits & number entry`

Primary: Digit 4

Keyboard: `4`

#44 5 `Digits & number entry`

Primary: Digit 5

Keyboard: `5`

#45 6 `Digits & number entry`

Primary: Digit 6

Keyboard: `6`

#46 × `Operators & execute`

Primary: Multiplication ×

Keyboard: `*` `X`

#47 ÷ `Operators & execute`

Primary: Division ÷

Keyboard: /

#48 1 `Digits & number entry`

Primary: Digit 1

Keyboard: 1

#49 2 `Digits & number entry`

Primary: Digit 2

Keyboard: 2

#50 3 `Digits & number entry`

Primary: Digit 3

Keyboard: 3

#51 + `Operators & execute`

Primary: Addition +

Keyboard: +

#52 − `Operators & execute`

Primary: Subtraction −

Keyboard: -

#53 0 Digits & number entry

Primary: Digit 0

Keyboard: 0

#54 • Digits & number entry

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

shifted (2nd / f / SHIFT layer) (press #1 first): π — π (3.14159265…) as an exact constant

Keyboard: .

#55 EXP Digits & number entry

Primary: ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

shifted (2nd / f / SHIFT layer) (press #1 first): e — Euler's number e (2.71828…)

Keyboard: e

#56 (−) Digits & number entry

Primary: Negative sign (−) — makes the next number negative. Different from the subtraction key

#57 Ans Memory, variables & registers

Primary: Ans — recall the last answer into the current expression

shifted (2nd / f / SHIFT layer) (press #1 first): HIST — history

5 · Worked examples

8 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Multiply two complex numbers (real part)

Calculate **Re((2+3i)(4−i))**. Press **Reset** first.

`#15` Re → `#32` (→ `#49` 2 → `#51` + → `#50` 3 → `#10` i → `#33`) → `#32` (→ `#43` 4 → `#52` − → `#10` i → `#33`) → `#33`) → `#7` EXE

Display shows `11` — expected 11. ✔ verified on the app.

Example 5.2 — Modulus of a complex number

Calculate **|3+4i|**. Press **Reset** first.

`#18` |z| → `#50` 3 → `#51` + → `#43` 4 → `#10` i → `#33`) → `#7` EXE

Display shows `5` — expected 5. ✔ verified on the app.

Example 5.3 — Imaginary part of a quotient

Calculate **Im((1+*i*)÷(1−*i*))**. Press **Reset** first.

`#16` Im → `#32` (→ `#48` 1 → `#51` + → `#10` *i* → `#33`) → `#47` ÷ → `#32` (→ `#48` 1 → `#52` − → `#10` *i* → `#33`) → `#33`) → `#7` EXE

Display shows `1` — expected 1. ✔ verified on the app.

Example 5.4 — Polar to rectangular (real part)

Calculate **Re(2∠60°)**. Press **Reset** first.

`#15` Re → `#49` 2 → `#11` ∠ → `#45` 6 → `#53` 0 → `#33`) → `#7` EXE

Display shows `1` — expected 1. ✔ verified on the app.

Example 5.5 — Argument in degrees

Calculate **Arg(1+*i*)**. Press **Reset** first.

`#17` Arg → `#48` 1 → `#51` + → `#10` *i* → `#33`) → `#7` EXE

Display shows `45` — expected 45. ✔ verified on the app.

Example 5.6 — Square of a complex number (imaginary part)

Calculate **Im((1+2*i*)²)**. Press **Reset** first.

`#16` Im → `#32` (→ `#48` 1 → `#51` + → `#49` 2 → `#10` *i* → `#33`) → `#20` x^2 → `#33`) → `#7` EXE

Display shows `4` — expected 4. ✔ verified on the app.

Example 5.7 — Parallel resistors

Calculate **6 ‖ 3 (Ω)**. Press **Reset** first.

`#45` 6 → `#29` ‖ → `#50` 3 → `#7` EXE

Display shows `2` — expected 2. ✔ verified on the app.

Example 5.8 — Gain in dB

Calculate **dB(10)**. Press **Reset** first.

`#25` dB → `#48` 1 → `#53` 0 → `#33`) → `#7` EXE

Display shows `20` — expected 20. ✔ verified on the app.

6 · Tricks & hacks

Hack 1: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 2: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = …

Hack 3: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 4: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 5: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 6: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 7: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 8: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 9: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 10: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 11: Hidden functions on this model

35 extra functions sit on shifted layers. Press the layer key once, then the key: for example #2 then #10 → X1; #2 then #11 → X2; #1 then #12 → ▸a+bi; #1 then #13 → CUBIC.

Hack 12: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #57.

Hack 13: Type faster on a computer

29 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 14: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 15: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 16: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
q	#20	x^2	Square x^2
r	#21	√	Square root √
^	#22	xy	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
n	#23	ln	Natural logarithm ln
l	#24	log	Common logarithm $\log_{10}$
s	#26	sin	Sine (uses the current angle unit)
c	#27	cos	Cosine (current angle unit)
t	#28	tan	Tangent (current angle unit)
(	#32	(	Opening bracket (
)	#33	)	Closing bracket). Closing brackets at the end of a line may be omitted
,	#34	,	Comma / argument separator (between function arguments, list items or coordinates)
7	#38	7	Digit 7
8	#39	8	Digit 8
9	#40	9	Digit 9
Backspace Delete	#41	DEL	DEL — delete the character at the cursor
Escape	#42	AC	AC — clear all of the current calculation
4	#43	4	Digit 4
5	#44	5	Digit 5
6	#45	6	Digit 6
* x	#46	×	Multiplication ×

/	#47	÷	Division ÷
1	#48	1	Digit 1
2	#49	2	Digit 2
3	#50	3	Digit 3
+	#51	+	Addition +
-	#52	–	Subtraction –
0	#53	0	Digit 0
.	#54	•	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
e	#55	EXP	×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 2: Wrong angle unit

If sin 30 gives –0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 3: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 4: Division precedence

$1 \div 2\pi$ may be read as $1 \div (2\pi)$ on textbook models. Type $1 \div (2 \times \pi)$ explicitly to be sure.

Mistake 5: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 6: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 7: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 8: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 9: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (8)

Work each one on calculator #1, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. Re((2+3i)(4−i)) = _____
2. |3+4i| = _____
3. Im((1+i)÷(1−i)) = _____
4. Re(2∠60°) = _____
5. Arg(1+i) = _____
6. Im((1+2i)²) = _____
7. 6 ‖ 3 (Ω) = _____
8. dB(10) = _____

Set B — Key-finding drills (82)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “SHIFT” (SHIFT / 2nd)? _____
2. Which key is “ALPHA” (ALPHA)? _____
3. Which key is “▲” (Cursor ▲)? _____
4. Which key is “▼” (Cursor ▼)? _____
5. Which key is “◀” (Cursor ◀)? _____
6. Which key is “▶” (Cursor ▶)? _____

7. **Which key is “EXE” (Execute / =)?** _____

8. **Which key is “SETUP” (SETUP)?** _____

9. **Which key is “ON” (ON)?** _____

10. **Which key is “i” (Imaginary unit i)?** _____

11. **Which key is “∠” (Polar entry r∠θ)?** _____

12. **Which key is “a+bi⇄” (Cycle result form: rectangular → polar → exponential)?** _____

13. **Which key is “QUAD” (Quadratic equation solver)?** _____

14. **Which key is “Conj” (Complex conjugate)?** _____

15. **Which key is “Re” (Real part)?** _____

16. **Which key is “Im” (Imaginary part)?** _____

17. **Which key is “Arg” (Argument)?** _____

18. **Which key is “|z|” (Modulus |z|)?** _____

19. **Which key is “$\sqrt[n]{\ }$ z” (All n complex n-th roots: $\sqrt[n]{\ }$Roots)?** _____

20. **Which key is “x^2” (Square x^2)?** _____

21. **Which key is “√” (Square root √)?** _____

22. **Which key is “xy” (Power x^y / ^)?** _____

23. **Which key is “ln” (Natural logarithm ln)?** _____

24. **Which key is “log” (Common logarithm $\log_{10}$)?** _____

25. **Which key is “dB” (Gain in decibels 20·log|z|)?** _____

26. **Which key is “sin” (Sine)?** _____

27. Which key is “cos” (Cosine)? _____

28. Which key is “tan” (Tangent)? _____

29. Which key is “||” (Parallel combination a||b = ab/)? _____

30. Which key is “ZC” (Capacitor impedance Zc)? _____

31. Which key is “DRG” (DRG)? _____

32. Which key is “(” (Opening bracket)? _____

33. Which key is “)” (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

34. Which key is “,” (Comma / argument separator)? _____

35. Which key is “STO” (STO)? _____

36. Which key is “RCL” (RCL)? _____

37. Which key is “M+” (M+)? _____

38. Which key is “DEL” (DEL)? _____

39. Which key is “AC” (AC)? _____

40. Which key is “×” (Multiplication ×)? _____

41. Which key is “÷” (Division ÷)? _____

42. Which key is “+” (Addition +)? _____

43. Which key is “–” (Subtraction –)? _____

44. Which key is “•” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

45. Which key is “EXP” (×10ˣ / EXP / EE)? _____

46. Which key is "(−)" (Negative sign)? _____

47. Which key is "Ans" (Ans)? _____

48. How do you reach "X1"? _____

49. How do you reach "X2"? _____

50. How do you reach "►a+bi"? _____

51. How do you reach "CUBIC"? _____

52. How do you reach "Sgn"? _____

53. How do you reach "A"? _____

54. How do you reach "Rot"? _____

55. How do you reach "B"? _____

56. How do you reach "►r∠θ"? _____

57. How do you reach "C"? _____

58. How do you reach "cis"? _____

59. How do you reach "D"? _____

60. How do you reach "Dist"? _____

61. How do you reach "E"? _____

62. How do you reach "CAT"? _____

63. How do you reach "F"? _____

64. How do you reach "x^3"? _____

65. How do you reach "$\sqrt[3]{}$"? _____

66. How do you reach “x^{-1}”? _____

67. How do you reach “ez”? _____

68. How do you reach “10x”? _____

69. How do you reach “n!”? _____

70. How do you reach “$\sin^{-1}$”? _____

71. How do you reach “sinh”? _____

72. How do you reach “$\cos^{-1}$”? _____

73. How do you reach “cosh”? _____

74. How do you reach “$\tan^{-1}$”? _____

75. How do you reach “tanh”? _____

76. How do you reach “ZL”? _____

77. How do you reach “CLRv”? _____

78. How do you reach “M−”? _____

79. How do you reach “M”? _____

80. How do you reach “π”? _____

81. How do you reach “e”? _____

82. How do you reach “HIST”? _____

Appendix A — Drill answers

Set A

1. 11 (display `11`)

#15 Re → #32 (→ #49 2 → #51 + → #50 3 → #10 *i* → #33) → #32 (→ #43 4 → #52 − → #10 *i* → #33) → #33) → #7 EXE

2. 5 (display `5`)

#18 |z| → #50 3 → #51 + → #43 4 → #10 *i* → #33) → #7 EXE

3. 1 (display `1`)

#16 Im → #32 (→ #48 1 → #51 + → #10 *i* → #33) → #47 ÷ → #32 (→ #48 1 → #52 − → #10 *i* → #33) → #33) → #7 EXE

4. 1 (display `1`)

#15 Re → #49 2 → #11 ∠ → #45 6 → #53 0 → #33) → #7 EXE

5. 45 (display `45`)

#17 Arg → #48 1 → #51 + → #10 *i* → #33) → #7 EXE

6. **4** (display `4`)

   `#16` lm → `#32` (→ `#48` 1 → `#51` + → `#49` 2 → `#10` *i* → `#33`) → `#20` x^2 → `#33`) → `#7` EXE

7. **2** (display `2`)

   `#45` 6 → `#29` ‖ → `#50` 3 → `#7` EXE

8. **20** (display `20`)

   `#25` dB → `#48` 1 → `#53` 0 → `#33`) → `#7` EXE

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #32
33. #33
34. #34
35. #35
36. #36
37. #37
38. #41
39. #42
40. #46
41. #47
42. #51
43. #52
44. #54
45. #55
46. #56
47. #57
48. #2 ALPHA then #10 i
49. #2 ALPHA then #11 ∠
50. #1 SHIFT then #12 a+bi⇄

51. #1 SHIFT then #13 QUAD
52. #1 SHIFT then #14 Conj
53. #2 ALPHA then #14 Conj
54. #1 SHIFT then #15 Re
55. #2 ALPHA then #15 Re
56. #1 SHIFT then #16 Im
57. #2 ALPHA then #16 Im
58. #1 SHIFT then #17 Arg
59. #2 ALPHA then #17 Arg
60. #1 SHIFT then #18 |z|
61. #2 ALPHA then #18 |z|
62. #1 SHIFT then #19 $\sqrt[n]{z}$
63. #2 ALPHA then #19 $\sqrt[n]{z}$
64. #1 SHIFT then #20 x^2
65. #1 SHIFT then #21 √
66. #1 SHIFT then #22 xy
67. #1 SHIFT then #23 ln
68. #1 SHIFT then #24 log
69. #1 SHIFT then #25 dB
70. #1 SHIFT then #26 sin
71. #2 ALPHA then #26 sin
72. #1 SHIFT then #27 cos
73. #2 ALPHA then #27 cos
74. #1 SHIFT then #28 tan
75. #2 ALPHA then #28 tan
76. #1 SHIFT then #30 ZC
77. #1 SHIFT then #31 DRG
78. #1 SHIFT then #37 M+
79. #2 ALPHA then #37 M+
80. #1 SHIFT then #54 •
81. #1 SHIFT then #55 EXP
82. #1 SHIFT then #57 Ans

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10^x / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^x 23	EXP	#55
10^x — antilog	10x	#1 → #24
AC — clear all of the current calculation	AC	#42
Addition +	+	#51
All n complex n-th roots: ⁿRoots(z,n)	ⁿ√z	#19
ALPHA — the next key types its letter/variable (red/green label)	ALPHA	#2
Ans — recall the last answer into the current expression	Ans	#57
Argument (angle) of z in the current unit	Arg	#17
Capacitor impedance Zc(C,f)	ZC	#30
Catalogue of complex functions	CAT	#1 → #19
cis(θ) = cos θ + i sin θ	cis	#1 → #17
Clear variables	CLRv	#1 → #31
Closing bracket). Closing brackets at the end of a line may be omitted	)	#33
Comma / argument separator (between function arguments, list items or coordinates)	,	#34
Common logarithm $\log_{10}$	log	#24
Complex conjugate	Conj	#14
cosh	cosh	#2 → #27
Cosine (current angle unit)	cos	#27
Cube root ∛	∛	#1 → #21

Cube x^3	x^3	#1 → #20
Cubic equation solver (complex roots)	CUBIC	#1 → #13
Cursor ▲ — up / previous history line / previous item	▲	#3
Cursor ▶ — move right in the line	▶	#6
Cursor ▼ — down / next history line / next item	▼	#4
Cursor ◀ — move left in the line	◀	#5
Cycle result form: rectangular → polar → exponential	a+bi⇄	#12
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	•	#54
DEL — delete the character at the cursor	DEL	#41
Digit 0	0	#53
Digit 1	1	#48
Digit 2	2	#49
Digit 3	3	#50
Digit 4	4	#43
Digit 5	5	#44
Digit 6	6	#45
Digit 7	7	#38
Digit 8	8	#39
Digit 9	9	#40
Distance between two complex points	Dist	#1 → #18
Division ÷	÷	#47
DRG — choose/convert the angle unit (Deg / Rad / Grad)	DRG	#31
Euler's number e (2.71828...)	e	#1 → #55
e^x — exponential	ez	#1 → #23

Execute / = — evaluate the line and show the result (EXE / ENTER / =)	EXE	#7
Factorial n! (also on non-integers via the gamma function where supported)	n!	#1 → #25
Gain in decibels 20·log|z|	dB	#25
history	HIST	#1 → #57
Imaginary part	Im	#16
Imaginary unit i	i	#10
Inductor impedance ZL(L,f)	ZL	#1 → #30
Inverse cosine $\cos^{-1}$	$\cos^{-1}$	#1 → #27
Inverse sine $\sin^{-1}$	$\sin^{-1}$	#1 → #26
Inverse tangent $\tan^{-1}$	$\tan^{-1}$	#1 → #28
M+ — add the value to independent memory M	M+	#37
M− — subtract the value from memory M	M−	#1 → #37
Modulus |z|	|z|	#18
Multiplication ×	×	#46
Natural logarithm ln	ln	#23
Negative sign (−) — makes the next number negative. Different from the subtraction key	(−)	#56
ON — power on	ON	#9
Opening bracket (	(	#32
Parallel combination a‖b = ab/(a+b)	‖	#29
Polar entry r∠θ	∠	#11
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	xy	#22
Quadratic equation solver (complex roots)	QUAD	#13
RCL — recall a memory / variable	RCL	#36

Real part	Re	#15
Reciprocal x^{-1} (1/x)	x^{-1}	#1 → #22
Rotate z by angle θ: Rot(z,θ)	Rot	#1 → #15
SETUP — angle unit, number format, display options	SETUP	#8
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#1
Show results in polar form	►r∠θ	#1 → #16
Show results in rectangular form	►a+bi	#1 → #12
Sine (uses the current angle unit)	sin	#26
sinh	sinh	#2 → #26
Square root √	√	#21
Square x^2	x^2	#20
STO — store the value into a memory / variable (press the target next)	STO	#35
Subtraction –	–	#52
Tangent (current angle unit)	tan	#28
tanh	tanh	#2 → #28
Unit phasor z/|z|	Sgn	#1 → #14
Variable A (store with STO/→, recall by name)	A	#2 → #14
Variable B (store with STO/→, recall by name)	B	#2 → #15
Variable C (store with STO/→, recall by name)	C	#2 → #16
Variable D (store with STO/→, recall by name)	D	#2 → #17

Variable E (store with STO/ →, recall by name)	**E**	**#2 → #18**
Variable F (store with STO/ →, recall by name)	**F**	**#2 → #19**
Variable M (store with STO/ →, recall by name)	**M**	**#2 → #37**
Variable X1	**X1**	**#2 → #10**
Variable X2	**X2**	**#2 → #11**
π (3.14159265…) as an exact constant	**π**	**#1 → #54**

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

Complex-number scientific for electrical engineering and algebra. Every value may be complex; results can be shown in rectangular, polar (r∠θ) or exponential form.

Trig functions of real arguments use the selected angle unit; complex arguments are always in radians (principal values).

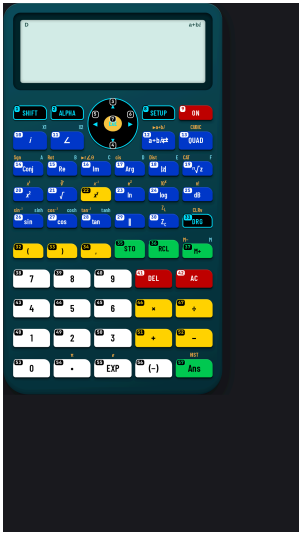
The logic is original U-45 code using the bundled open-source math.js library (Apache-2.0).

Planned (not yet working)

- **Exam mode (locks memory & shows indicator)**
- **Answer history export to text**
- **Speech read-out of results**
- **Colour-blind safe key theme**
- **Custom shortcut keys**

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 SHIFT	#2 ALPHA	#3 ▲
#4 ▼	#5 ◀	#6 ▶
#7 EXE	#8 SETUP	#9 ON
#10 i	#11 ∠	#12 a+bi⇄ ⇧ ▸a+bi
#13 QUAD ⇧ CUBIC	#14 Conj ⇧ Sgn	#15 Re ⇧ Rot
#16 Im ⇧ ▸r∠θ	#17 Arg ⇧ cis	#18 |z| ⇧ Dist
#19 $\sqrt[n]{z}$ ⇧ CAT	#20 x^2 ⇧ x^3	#21 $\sqrt{}$ ⇧ $\sqrt[3]{}$
#22 xy ⇧ x^{-1}	#23 ln ⇧ e^x	#24 log ⇧ 10^x
#25 dB ⇧ n!	#26 sin ⇧ $\sin^{-1}$	#27 cos ⇧ $\cos^{-1}$
#28 tan ⇧ $\tan^{-1}$	#29 ‖	#30 ZC ⇧ ZL
#31 DRG ⇧ CLRv	#32 (	#33)
#34 ,	#35 STO	#36 RCL
#37 M+ ⇧ M−	#38 7	#39 8
#40 9	#41 DEL	#42 AC
#43 4	#44 5	#45 6
#46 ×	#47 ÷	#48 1
#49 2	#50 3	#51 +

#52 –	**#53** 0	**#54** • ⇧ π
#55 EXP ⇧ e	**#56** (–)	**#57** Ans ⇧ HIST

U-45 CALCULATORS · Calculator #1 · cut out and keep